

Sample House, Ground Floor Dwelling

Doc: ENE-201 Rev: B Issued: 2026-05-20 By: Fjordsyn Rådgivende Ingeniører AS

Thermal Specification

1. Purpose

This specification sets the thermal transmittance (U-value) limits for the envelope elements of the dwelling in `Ifc4_SampleHouse.ifc`, for verification against TEK17 chapter 14 (energy) using the component method, §14-2 second paragraph. Values are in $\text{W/m}^2\text{K}$. Element types are referred to by their model Reference property; see the schedule in §4.

Revision B corrects the ground floor limit after the soil thermal resistance review (see §3.4).

2. Scope limits

Thermal bridging, airtightness and ventilation heat recovery are covered in ENE-202 and are not part of this document. Values stated as "as designed" in §3 are recorded for the energy model and carry no requirement.

3. U-value requirements

3.1 External walls

External cavity walls (Wall-Ext_102Bwk-75Ins-100LBlk-12P) shall have a thermal transmittance not exceeding $0.18 \text{ W/m}^2\text{K}$. The current build-up with 75 mm cavity insulation is not expected to meet this without an upgraded insulation specification; the energy model shall use the required value, not the as-modelled value.

3.2 Windows

Windows (Windows_Sgl_Plain 1810x1210mm) shall have a whole-window U-value not exceeding $1.2 \text{ W/m}^2\text{K}$, corresponding to a double-glazed low-E unit with argon fill. Single glazing is not acceptable.

3.3 Roof

The flat roof (Roof_Flat-4Felt-150Ins-50Scr-150Conc-12Plr) shall have a thermal transmittance not exceeding $0.13 \text{ W/m}^2\text{K}$.

3.4 Ground floor

The suspended ground floor (Floor-Grnd-Susp_65Scr-80Ins-100Blk-75PC) shall have a thermal transmittance not exceeding $0.10 \text{ W/m}^2\text{K}$, ground resistance included. Revision A stated 0.15; the tighter value is required to offset the wall shortfall in the energy balance.

3.5 External door

The external double door (Doors_ExtDbl_Flush 1810x2110mm) shall have a U-value not exceeding $1.2 \text{ W/m}^2\text{K}$. The model does not currently carry a thermal transmittance value for this door; the supplier declaration shall be entered into the model before the energy calculation is finalised.

3.6 Internal doors and partitions (as designed, no requirement)

Internal doors (Doors_IntSgl 810x2110mm) are recorded at $3.7 \text{ W/m}^2\text{K}$ as designed. Internal stud partitions

(Wall-Partn_12P-70MStd-12P) are recorded at 0.35 W/m²K as designed. Both separate heated spaces and carry no U-value requirement under TEK17 §14-2.

The roof-level access deck slab ("Simple floor") is outside the thermal envelope and carries no requirement.

4. Schedule of limits

Element	Model reference	U-value limit (W/m ² K)
External wall	Wall-Ext_102Bwk-75Ins-100LBlk-12P	≤ 0.18
Window	Windows_Sgl_Plain 1810x1210mm	≤ 1.2
Roof	Roof_Flat-4Felt-150Ins-50Scr-150Conc-12Plr	≤ 0.13
Ground floor	Floor-Grnd-Susp_65Scr-80Ins-100Blk-75PC	≤ 0.10
External door	Doors_ExtDbl_Flush 1810x2110mm	≤ 1.2
Internal door	Doors_IntSgl 810x2110mm	as designed, 3.7
Internal partition	Wall-Partn_12P-70MStd-12P	as designed, 0.35

5. References

- TEK17 Byggteknisk forskrift, kapittel 14 Energi, §14-2
- NS 3031:2014 Calculation of energy performance of buildings
- NS-EN ISO 6946 Building components, thermal resistance and transmittance
- NS-EN ISO 13370 Heat transfer via the ground